BPhO Computational Challenge 2026

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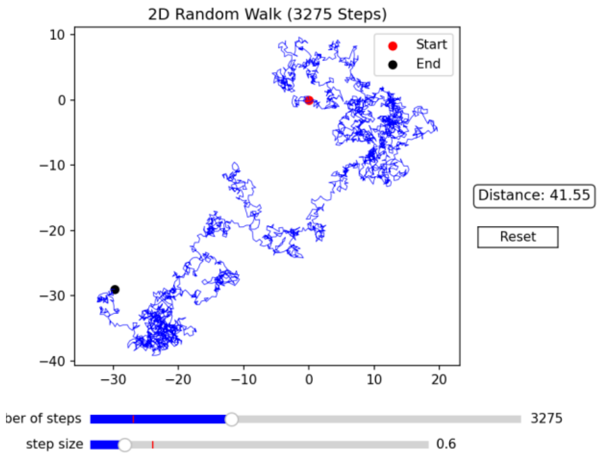
Abstract

In this document, we are outlining how we are modelling each task that has been given in the computational challenge, where we explain the theory and the formulae behind each model. In terms of the coding language that we used, all of the models were locally ran in python, using NumPy and SciPy for the mathematics of the model and for visualisations, used Matplolib, although task 3 used VPython instead.

Task #1. Random Walk

A random walk is a discrete stochastic process where the particle moves randomly every step. In this model, the particle moves in a random direction from 0 to 2pi radians every step.

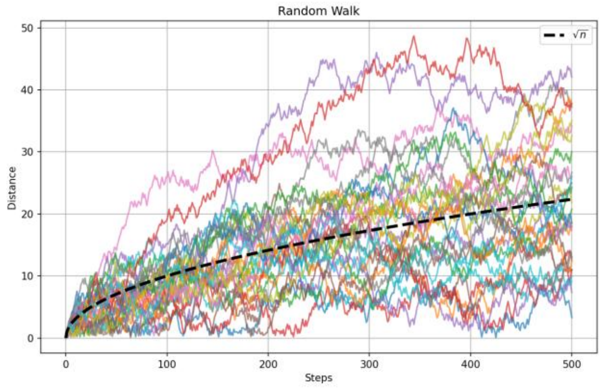


Random walks follow the square root law, where the expected distance (or Root Mean Square displacement, R_{rms}) after N steps is calculated as the square root of the number of steps multiplied by the length of the step.

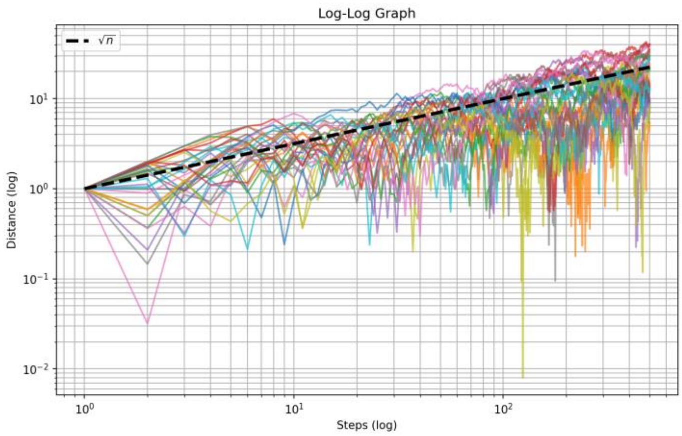
$$R_{rms} = L\sqrt{N}$$

In this context, we made the step size to be one, which makes it inconsequential to the expectation of the distance, and so if we have a certain number of steps, the expected distance of the particle from the origin should be $\sqrt{N}$.

We decided to model this random walk 30 times, stopping after 500 iterations, and then looked at the shape of the distance against the number of steps, they do seem to take a similar shape to a square root graph.



Again, taking a log-log graph of the data, the pattern is clear, although the graph does look quite chaotic.



Task #2. Brownian Motion

Brownian motion is defined as the random movement of a smoke particle (or another other comparable particle) as it collides with other randomly moving particles. As the movement of the particle in Brownian motion is to be random, the motion that it follows can be comparable to that of a random walk model, and so in theory should follow the same patterns as that of a random walk model.

How this is going to be model is going to be via a larger "smoke" particle that has a different colour to hundreds of other smaller particles both in volume and in mass. While in theory the smaller particles would be

Another factor to consider is how the collisions are going to work between particles. For the sake of an ideal model, we have decided that the particles will have perfectly elastic collisions with each other, conserving energy and momentum in each collision. So, for example, if a there is a particle with mass $m_1$and velocity u_2 which collides with another particle with mass m_2 and velocity u_2: their kinetic energy will be conserved such that their final velocities v_1 and v_2 will follow the formulae:

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

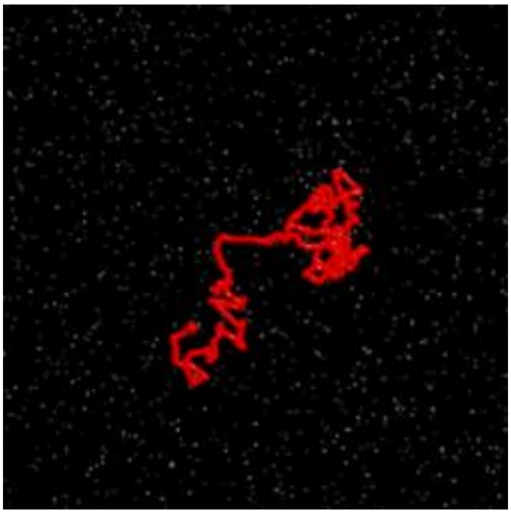
Rearranging and solving for v_1 and v_2 we can reach the following equations:

$$v_1 = \frac{(m_1 - m_2)u_1 + 2m_2u_2}{m_1 + m_2}$$

$$v_1 = \frac{(m_2 - m_1)u_2 + 2m_1u_1}{m_1 + m_2}$$

They direction at which the collided particles move is calculated as tangential to the vector between the two particles at the point of collision

(It is also important to note that the smaller particles will be moving in a random walk model and will only be colliding with the larger particle as has been instructed in the task as to not make the model difficult to run, and it should not make an effect on the model as there are so many smaller particles that exist).



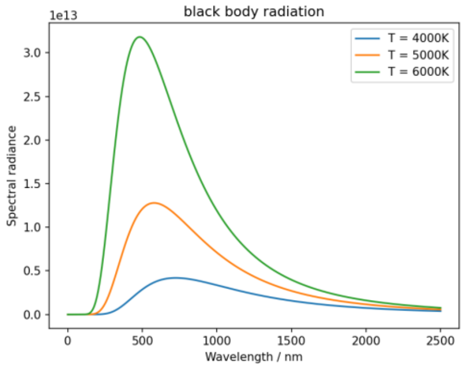
Unlike other models, we coded this one in VPython, as we deemed it much easier than doing it on matplotlib or any other python module.

Task #3. Plank "Black Body Radiation" Spectrum

A black body is an ideal object that absorbs all electromagnetic radiation that hits it and emits radiation depending only on its temperature. If a black body is heated until is glows, Planck's black body radiation spectrum determines how much energy is emitted at each wavelength. Planck's law displays the spectral radiance $B(\lambda, T)$ (measured in $W \cdot sr^{-1} \cdot m^{-3}$ or watts per steradian per metre cubed where a steradian is the standard scientific unit used to measure solid angles in three-dimensional space).

$$B(\lambda, T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{\frac{hc}{\lambda k_B T}} - 1}$$

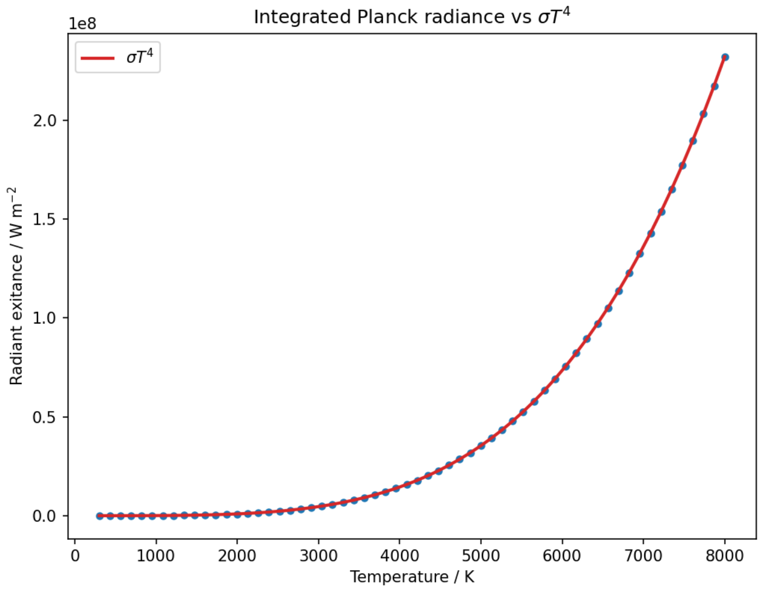
Where h is Planck's constant and k_B is Boltzmann's constant.



It is also determined that the total energy that is emitted by the black body is equal to σT^4 where σ is the Stefan-Boltzmann constant. Therefore, the integral of $B(\lambda, T)$ from 0 to infinity must equal to σT^4.

$$\int_0^\infty B(\lambda, T)\, d\lambda = \sigma T^4$$

To verify this, we numerically integrated the spectra at different values of T and also plotted σT^4 alongside to see if it was a match. As is evident by the graph below, σT^4 certainly does equal to the total radiance emitted by the black body, with differences being due to the error from the numerical integration or not being able to integrate up to infinity.

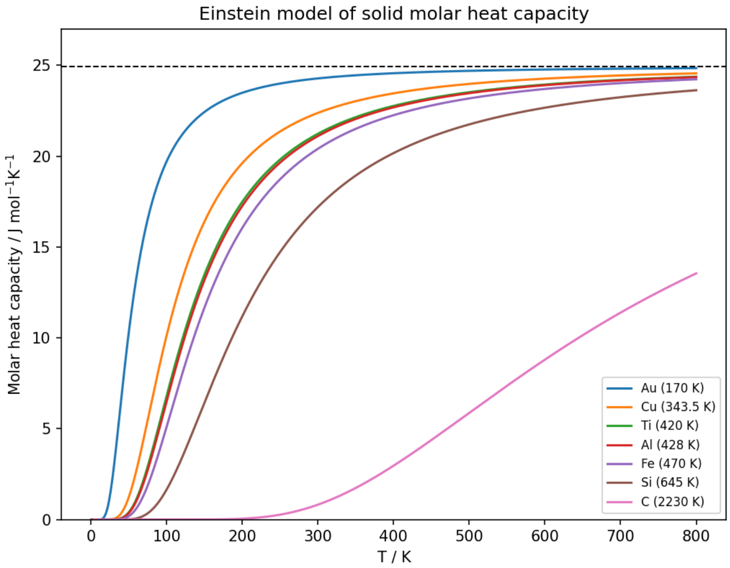


Einstein's model of molar heat capacity uses Planck's black body radiation spectrum to describe the effects of temperature on the heat capacity of solids. The model assumes all atoms in a crystalline solid act as independent quantum harmonic oscillators that vibrate at the exact same frequency, which led to the formula of Einstein's model of molar heat capacity being the following.

$$C = 3R\frac{x^2 e^x}{(e^x - 1)^2}$$

$$x = \frac{hf_E}{k_B T}$$

We can then plot this model, which shows the Dulong and Petit limit of $3R$.



Task #4. Photoelectric Effect

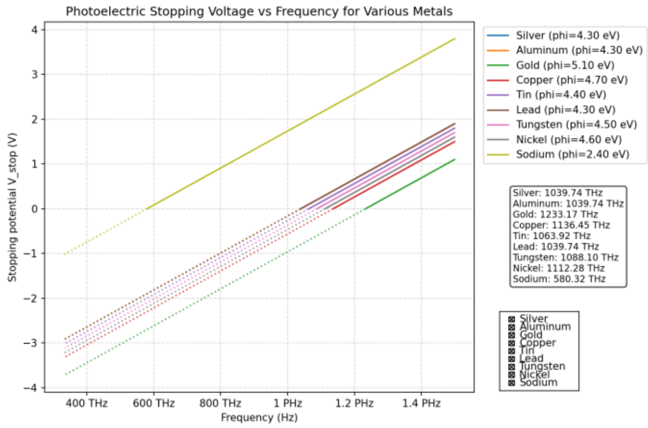
The photoelectric effect is a phenomenon in physics that led to the theory that photons exist as quantised particles as well as waves. It involves shining UV light on a metal, where the metal releases photoelectrons as a result.

The intensity of the light hitting the metal did not have any bearing on whether there are photoelectrons that are released, and instead it depends on the frequency of the photon. This is because light is comprised of packets, photons, and only a single photon is able to release an electron. A photon's energy is given by the following equation:

$$E = hf$$

If the photon's energy E is greater than or equal to the electron's work function ϕ, then all of the photon's energy is transferred to the electron, with any excess energy being transferred into kinetic energy. Therefore, the kinetic energy that is released as a result of the photoelectric effect can be described as the difference between the energy of the photon and the work function of the material.

$$KE_{max} = hf - \phi$$



TASK #5. Hydrogen emission spectrum and Bohr's model of a hydrogenic atom.

In Bohr's model of a hydrogen atom, electrons exist in discrete quantised energy levels, which range from n = 1 to infinity. As the energy levels get higher, the gaps between each level gets smaller and tends to 0. The energy of each level is defined by the formula:

$$E_n = -\frac{13.6Z^2}{n^2}eV$$

Where Z is the number of electrons orbiting the nucleus, which in this case is 1 and therefore would not have a consequence on the formula.

When an electron moves from a higher energy level to a lower energy level, energy is released via a photon and due to conservation of energy the energy of the photon will equal

$$E_{photon} = E_{higher} - E_{lower}$$

And since

$$E = f\lambda$$

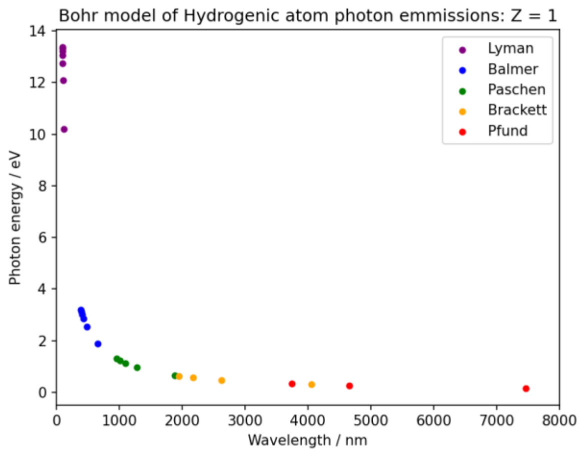
$$E = \frac{hc}{\lambda}$$

We can find the relationship between the energy of the photon and the wavelength of it as being an inverse relationship

$$\lambda = \frac{hc}{E_{photon}}$$

The transitions are categorised by the destination of the electron, where every single transition that lands on the energy level n = 1 is in the Lyman series, where the photon's wavelengths result in them being UV light. Any transition that lands on the energy level n = 2 is in the Balmer series which is visible light and is the transition that one would find the wavelength of the flame test, and any other. Transitions that land on n = 3 is then infrared.

Modelling helps with depicting the data much easier as it is able to manually calculate the energy of each transition, which in this case we did for the highest transition being from the eighth energy level, and then can calculate the resulting wavelength of the photon to display the graph of Bohr's model of the hydrogen atom.



Task #6. Electron Diffraction

When an electron passes through graphite which is made up of stacked hexagonal carbon layers, it diffracts in the same way as when light passes through a diffraction grating. According to Bragg's law, two waves reflecting off adjacent layers travel slightly different path lengths, where the path length is equal to $2d \sin \phi$, where ϕ is the angle between the incoming beam and the crystal plane, known and the Bragg angle.

As a result, using the formula for the pattern of a diffraction grating, where the path difference must be at multiples of the wavelength for there to be constructive interference, the following formula, which is Bragg's law, must be true.

$$n\lambda = 2d \sin \phi$$

Moreover, we know that electrons have an associated de Broglie wavelength calculated via the momentum p, which is also equal to mv.

$$\lambda = \frac{h}{p}$$

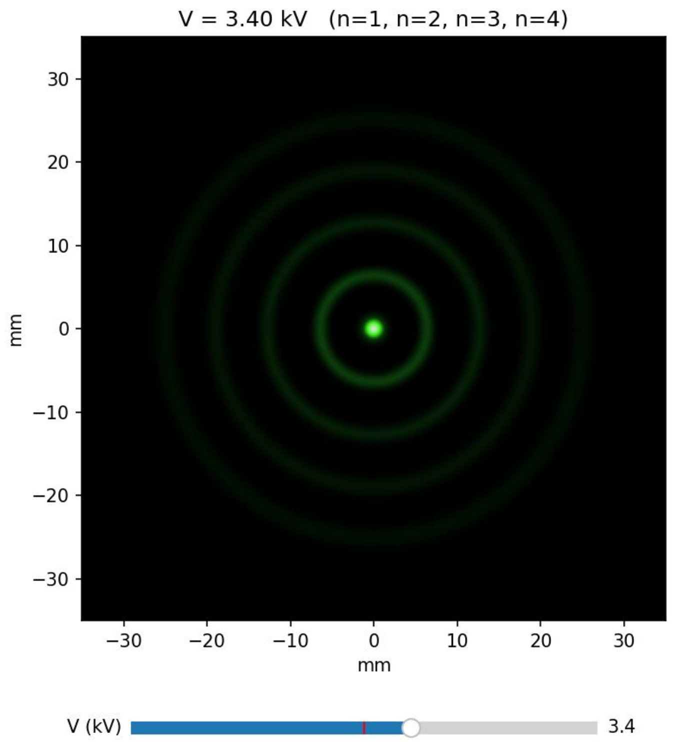
Since $p = \sqrt{2meV}$, the aforementioned can be substituted into the above equation, and then into Bragg's law to arrive to the conclusion that $\sin \phi$ must fulfil the following.

$$\sin \phi = \frac{n\lambda}{2d} = \frac{nh}{2d\sqrt{2meV}}$$

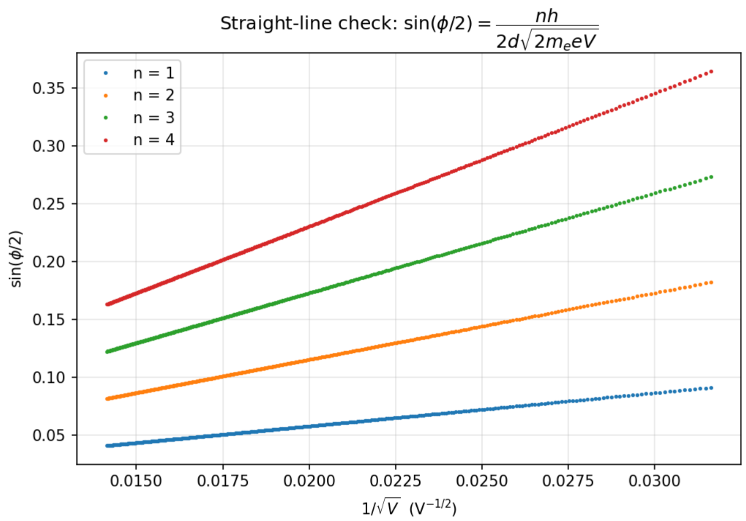
And so, the diffraction angle ϕ is dependent on V.

As the diffracted wave comes in from a point that is subtended on the circumference, the electron creates a cone of half angle 2ϕ, where the tip of the cone is at the centre of the sphere as an angle subtended at the centre is double that of an angle subtended at the circumference, so the pattern creates a ring where $x = r \sin 2\phi$. Therefore, in this context as opposed to $\sin \phi$, the following equation describes the shape of the rings:

$$\sin\left(\frac{\phi}{2}\right) = \frac{nh}{2d\sqrt{2meV}}$$



To verify the model, we can plot a graph of $\sin\left(\frac{\phi}{2}\right)$ against $\frac{1}{\sqrt{V}}$ and see whether it is a straight line. The method in which this can be done is through a linear regression model, calculating the r^2 value of the point and as a result verifying it.



Task #7. Particle-in-a-box solution to the Schrödinger Equation

The particle-in-a-box experiment describes an electron trapped inside of a box, where the probability of the position within the box is described by the wavefunction $\psi(x)$ and the probability density of the electron's position at any point x is described as the absolute squared value of the wavefunction, $|\psi(x)|^2$.

Inside of the box, the walls are infinitely high, and so the particle cannot ever be found outside of the box, therefore arriving to the boundary condition such that, with a box of length L,

$$\psi(0) = \psi(L) = 0$$

The way in which the wavefunction of the electron is described in this environment is through the time-independent Schrödinger equation:

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} = E\psi$$

Rearranging the equation, it can take the form of a homogeneous second-order linear ODE that can be solved via setting up the auxiliary equation:

$$\psi'' = -\frac{2mE}{\hbar^2}\psi$$

By letting $\frac{2mE}{\hbar^2} = k^2$, we can find the general solution of the differential equation to be:

$$\psi(x) = A\sin(kx) + B\cos(kx)$$

Applying the boundary conditions:

$$\psi(0) = 0 \Longrightarrow B = 0$$

$$\psi(L) = 0 \Longrightarrow A\sin(kL) = 0 \Longrightarrow kL = n\pi, n \in \mathbb{N}$$

Since $\psi'' = -k^2\psi$, plugging that back into the differential equation, and knowing that $\hbar = h/2\pi$ we can find that

$$E_n = \frac{\hbar^2 k^2}{2m} = \frac{n^2\pi^2\hbar^2}{2mL^2} = \frac{n^2 j^2}{8mL^2}$$

This is the formula which is then used to calculate the energy of the particle.

Knowing that $\psi_n(x) = \mathrm{A}\sin\left(\frac{n\pi x}{L}\right)$, and that the probability density of the particle is going to total 1 when taking the integral of the wavefunction across the entirety of the box. Therefore, the following integral must hold true the following condition, allowing us to find the value of A.

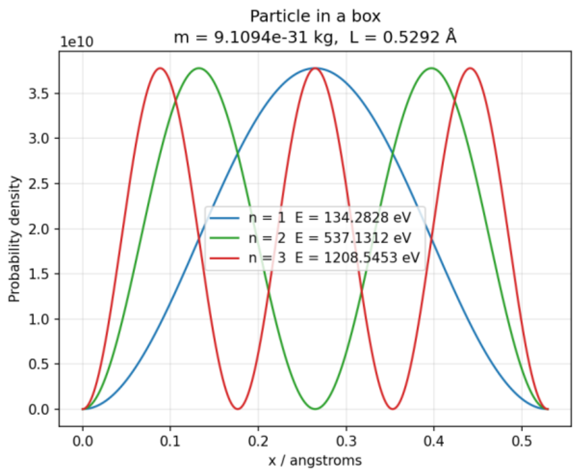
$$\int_0^L |\psi_n(x)|^2\, dx = 1 \Longrightarrow A^2 \int_0^L sin^2\left(\frac{n\pi x}{L}\right) dx = 1$$

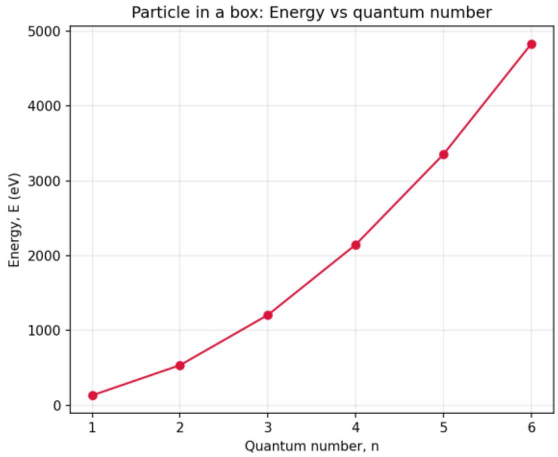
Solving for A, we can find that the value of A is equal to $\sqrt{2/L}$. Thus, the wavefunction for a particle-in-a-box is:

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)$$

And so, the probability density at any point x will be:

$$P_n(x) = |\psi_n(x)|^2 = \frac{2}{L}sin^2\left(\frac{n\pi x}{L}\right)$$





The uncertainty principle is where $\Delta x \Delta p \geq \frac{1}{2}\hbar$, where Δx is the uncertainty of the position operator of the particle and Δp is the uncertainty of the momentum operator of the particle. These operators $\hat{x}$ and $\hat{p}$ are defined through the following formulae.

$$\hat{x}\psi = x\psi, \hat{p}\psi = -i\hbar\frac{d\psi}{dx}$$

The uncertainty for $\hat{x}$, Δx is calculated via calculating the standard deviation of the expectation of the operator of the particle's position. The same can be done for the momentum of the particle.

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$$

Where the expectation of the position of the particle, $\langle x \rangle$, is calculated through the following integral, which is also used to calculate the expectation of continuous variables in statistics.

$$\langle x \rangle = \int_0^L x|\psi_n(x)|^2 \, dx$$

Therefore, $\langle x \rangle$, $\langle x^2 \rangle$, $\langle p \rangle$, $\langle p^2 \rangle$ can be found be computing the integrals.

$$\langle x \rangle = \frac{2}{L}\int_0^L x sin^2\left(\frac{n\pi x}{L}\right) dx = \frac{L}{2}$$

$$\langle x^2 \rangle = \frac{2}{L}\int_0^L x^2 sin^2\left(\frac{n\pi x}{L}\right) dx = \frac{L^2}{3} - \frac{L^2}{2n^2\pi^2}$$

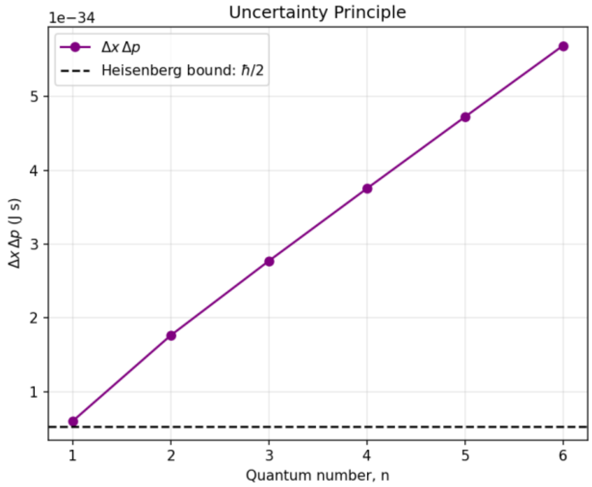
$$\langle p \rangle = \int_0^L \psi_n\left(-i\hbar\frac{d\psi_n}{dx}\right) dx = 0$$

$$\langle p^2 \rangle = \int_0^L \psi_n\left(-i\hbar\frac{d\psi_n}{dx}\right)^2 dx = \frac{n\pi\hbar}{L}$$

Using these values, Δx and Δp by plugging it into the formula.

$$\Delta x = L\sqrt{\frac{1}{12} - \frac{1}{2n^2\pi^2}}$$

$$\Delta p = \frac{n\pi\hbar}{L}$$



As is evident by the shape of the graph, all values of $\Delta x \Delta p$ is larger than $\frac{1}{2}\hbar$, with the values of it being strictly increasing, approaching larger values for larger values of n.

Task #8. Quantum Cryptography

Quantum cryptography is an experiment in which the classical model of physics and the quantum model of physics both have different models in calculating probabilities due to the nature of entangled particles. The experiment involves two entangled photons travelling in two opposite directions where there is a detector in each direction, with both detectors having a polarising filter in front of them, rotated by an angle θ and φ respectively. Whether the photon passes through the filter and is detected is the result of the experiment, where the experiment is measuring the probability that there is a mismatch between the two photons, i.e. one photon passes through while the other does not.

Since a photon is measured as a wave, it can be oscillating in any plane of an angle θ. In a non-probabilistic, classical physical model, the resulting intensity of a polarised ray of light transmitting through a filter can be modelled through Malus' Law:

$$E_\theta = E_0 cos^2(\theta)$$

The same can be done for a single photon passing through a polarising filter, instead describing the probability that it passes through, where:

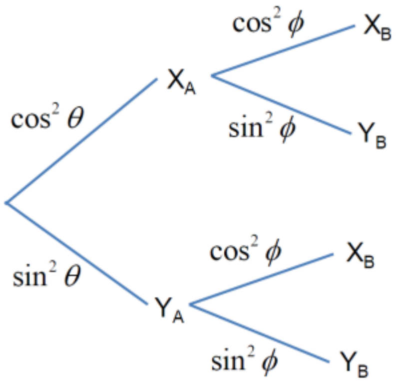
$$P(pass) = cos^2(\theta)$$

As a result, since the assumption is that the entangled photons are polarised in the same direction and plane, assuming that there are two planes X_A and Y_A, the probability of the particle being detected as X_A is the following:

$$P(X_A) = cos^2\theta$$

$$P(Y_A) = P'(X_A) = sin^2\theta$$

In the context of a classical model, the probabilities of X_B and Y_B are considered to be independent from the outcome of the detector A, and therefore the probabilities can be expressed via this tree diagram:



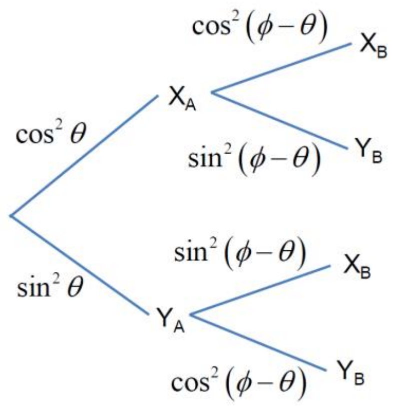
Using this tree diagram, the probabilities of match or a mismatch can be calculated.

$$P(match) = P(X_A, X_B) + P(Y_A, Y_B)$$

$$P(match) = cos^2\,\theta\, cos^2\,\phi + sin^2\,\theta\, sin^2\,\phi$$

$$P(mismatch) = P'(match) = 1 - cos^2\,\theta\, cos^2\,\phi + sin^2\,\theta\, sin^2\,\phi$$

However, in a quantum system, which assumes the dependence of the photons, where they are entangled and therefore B is definitely aligned along the same physical direction that A's measurement just found. Therefore, the probability that X_B is detected is now $cos^2(\phi - \theta)$. And so, the probabilities of each permutation can be expressed by the tree diagram.



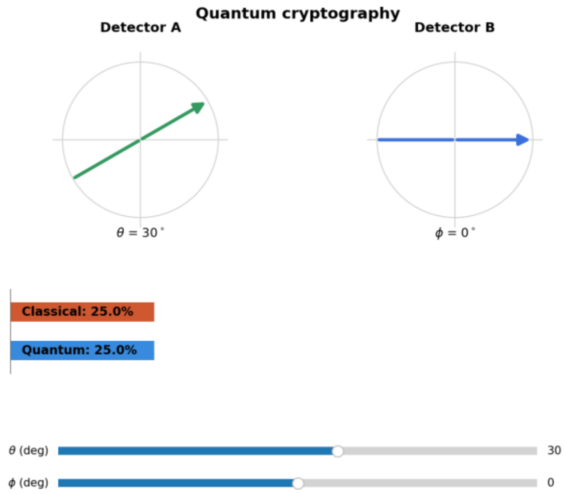
By consulting this tree diagram, the following conclusions can be made for the probability of a match of a mismatch.

$$P(match) = P(X_A, X_B) + P(Y_A, Y_B)$$

$$P(match) = cos^2\,\theta\, cos^2(\phi - \theta) + sin^2\,\theta\, cos^2(\phi - \theta)$$

$$P(match) = cos^2(\phi - \theta)$$

$$P(mismatch) = P(mismatch) = P'(match) = sin^2(\phi - \theta)$$



Task #9. Compton Scattering

In Compton scattering, the Compton effect refers to the scattering of X-rays of free electrons. This is modelled as a two-dimensional collision between an X-ray photon of wavelength λ and a free electron of mass m_e. Since the photon's wavelength can be calculated by rearranging the formula for $E = hf$ such that we have a formula of λ in terms of E.

$$\lambda = \frac{hc}{E_\lambda}$$

After the collision, the photon moves in a direction of angle θ while the electron moves in a direction of angle ϕ. In the collision, the energy and the momentum must be conserved before and after the collision. Therefore, since the following equation must be true, the Compton Shift equation can be derived from that idea.

$$\frac{hc}{\lambda} + m_e c^2 = \frac{hc}{\lambda'} + E_e$$

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta)$$

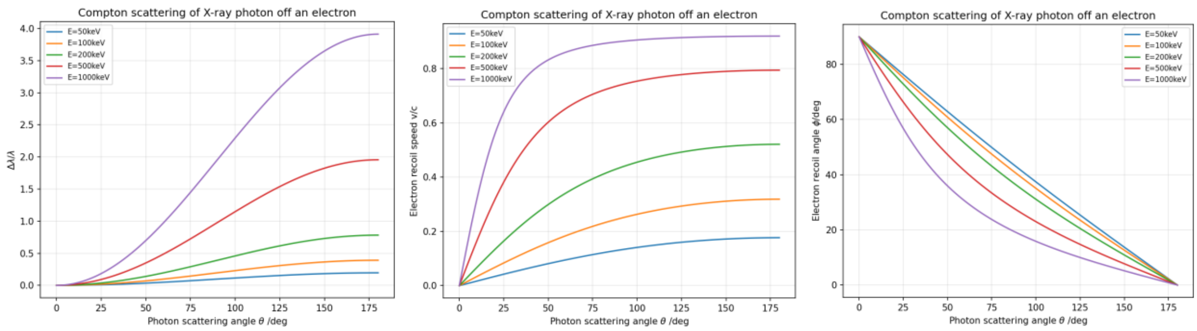
The recoil formula, again based upon the fact that energy and momentum are conserved in the collision, allows for the calculation of the new velocity of the photon.

$$v = c\sqrt{1 - \left(\frac{m_e c^2}{\frac{hc}{\lambda} - \frac{hc}{\lambda'} + m_e c^2}\right)^2}$$

Finally, the formula which dictates the angle at which the electron moves is with the recoil angle formula, which is described as the following:

$$\tan\phi = \frac{\sin\theta}{1 + \frac{h}{m_e c}(1 - \cos\theta) - \cos\theta}$$

We can then plot all of these equations, with different values of energy.



Task #10. Hydrogenic Orbitals

To determine what possible wavefunctions could exist in an orbital, the Schrödinger equation is a partial differential equation that describes the probability that an electron could exist in certain areas of an atom, given n, l and m quantum numbers. The equation is as follows.

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\psi$$

n is the principal quantum number where $n \in \mathbb{N}$, l is the electron's orbital angular momentum, where it is always equal to $n-1$, and m is the quantum number related to the orientation of the angular momentum relative to a chosen axis, where m can be any integer from $-l$ to l, thereby being $2l+1$ different possibilities of values of m for any value of l.

The probability density of the electron's location is proportional to the absolute squared of the probability amplitude, which is given by the wavefunction, and so the probability density such that an electron can be found in a position is as follows.

$$P(x,y,z) \propto |\psi(x,y,z)|^2$$

For a hydrogen atom, the wavefunction can be solved in polar coordinates and can be split into two parts, where one represents the radial effect and the other the angular effect on the wavefunction.

$$\psi(r,\theta,\phi) = R_{nl}(r)Y_l^m(\theta,\phi)$$

Therefore, the probability density of the electron's location in terms of polar coordinates is the radial wavefunction squared multiplied by the angular wavefunction.

$$|\psi|^2 = |R(r)|^2|Y(\theta,\phi)|^2$$

The radial wavefunction for the hydrogenic atom is given by the formula:

$$R_{nl}(r) = \sqrt{\left(\frac{2Z}{na_0}\right)^3\frac{(n-l-1)!}{2n[(n+l)!]}}\,e^{-\frac{Zr}{na_0}}\left(\frac{2Zr}{na_0}\right)^l L_{n-l-1}^{2l+1}\left(\frac{2Zr}{na_0}\right)$$

Where $L_{n-l-1}^{2l+1}\left(\frac{2Zr}{na_0}\right)$ is an associated Laguerre polynomial and the exponential decay term indicates that as the value of r tends to higher numbers, the probability becomes increasingly less likely. A Laguerre polynomial is defined as the sum:

$$L_n^{\alpha}(x) = \sum_{k=0}^{n} (-1)^k \frac{\Gamma(n+\alpha+1)}{\Gamma(k+\alpha+1)(n-k)!\,k!} x^k$$

The angular wavefunction for the hydrogenic atom is given by the formula:

$$Y_l^m(\theta,\phi) = \sqrt{\frac{2l+1}{4\pi}\frac{(l-|m|)!}{(l+|m|)!}} P_l^{|m|}(\cos\theta) e^{im\phi}$$

Where $P_l^{|m|}(\cos\theta)$ is an associated Legendre function, where a Legendre function is the solution to the following differential equation:

$$\frac{d}{dx}\left[(1-x^2)\frac{dy}{dx}\right] + \left[l(l+1) - \frac{m^2}{1-x^2}\right] y = 0$$

$$y = P_l^m(x)$$

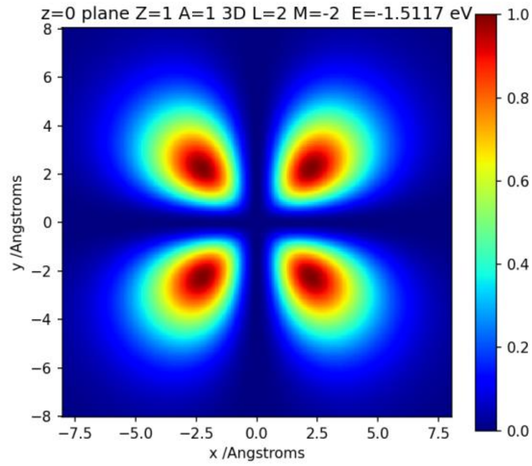
The probability of finding an electron in a certain volume V can be given as the sum of all probability densities in that volume or expressed as the following integral.

$$P = \int_V |\psi|^2 \, dV$$

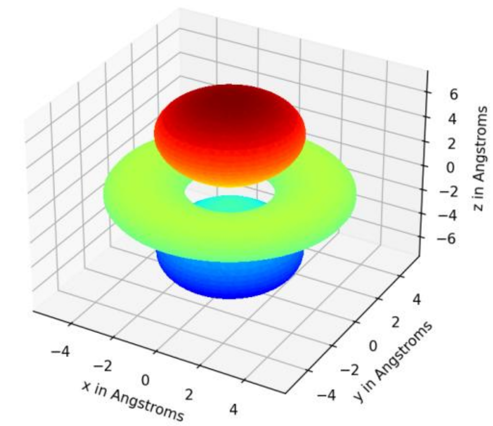
And since this is a probability function, the integral of the entire volume must equal to 1.

$$\int_{all\ volume} |\psi|^2 \, dV = 1$$

To model the 2D slice, it is in the xy plane where $z = 0$. Therefore, by fixing the value of ϕ to be 0 in the angular wavefunction, the 2D slice of the hydrogenic atom can be visualised.



The below is the 3D visualisation of the hydrogenic atom for $n = 1, l = 2, m = -2$



References

Apart from the resources given to us by those running the computational challenge, to help understand the theory behind the models we also used the following sources.

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